

2.2 APB Protocol

Advanced Peripheral Bus (APB) is the part of Advanced Microcontroller Bus Architecture (AMBA) family protocols. The latest version of APB is v2.0, which was a part of AMBA 4 release. It is a low-cost interface and it is optimized for minimal power consumption and reduced interface complexity. Unlike AHB, it is a Non-Pipelined protocol, used to connect low-bandwidth peripherals. Mostly, used to connect the external peripheral to the SOC. In APB, every transfer takes at least two clock cycles (SETUP Cycle and ACCESS Cycle) to complete. It can also interface with AHB and AXI protocols using the bridges in between.

Table 11 APB Pin Description

Signal	Source	Description
PCLK	Clock Source	Clock. The rising edge of PCLK times all transfers on the APB.
PRESETn	System Bus Equivalent	Reset. The APB reset signal is active LOW. This signal is normally connected directly to the system bus reset signal
PADDR	APB Bridge	Address. This is the APB address bus. It can be up to 32 bits wide and is driven by the peripheral bus bridge unit.
PROT	APB bridge	Protection type. This signal indicates the normal, privileged, or secure protection level of the transaction and whether the transaction is a data access or an instruction access
PSELx	APB bridge	Select. The APB bridge unit generates this signal to each peripheral bus slave. It indicates that the slave device is selected and that a data transfer is required. There is a PSELx signal for each slave.
PENABLE	APB bridge	Enable. This signal indicates the second and subsequent cycles of an APB transfer
PWRITE	APB bridge	Direction. This signal indicates an APB write access when HIGH and an APB read access when LOW.
PWDATA	APB bridge	Write data. This bus is driven by the peripheral bus bridge unit during write cycles when PWRITE is HIGH. This bus can be up to 32 bits wide.

PSTRB	APB bridge	Write strobes. This signal indicates which byte lanes to update during a write transfer. There is one write strobe for each eight bits of the write data bus. Therefore, PSTRB[n] corresponds to PWDATA[(8n + 7):(8n)]. Write strobes must not be active during a read transfer.
PREADY	Slave interface	Ready. The slave uses this signal to extend an APB transfer
PRDATA	Slave interface	Read Data. The selected slave drives this bus during read cycles when PWRITE is LOW. This bus can be up to 32-bits wide.
PSLVERR	Slave interface	This signal indicates a transfer failure. APB peripherals are not required to support the PSLVERR pin. This is true for both existing and new APB peripheral designs. Where a peripheral does not include this pin then the appropriate input to the APB bridge is tied LOW.

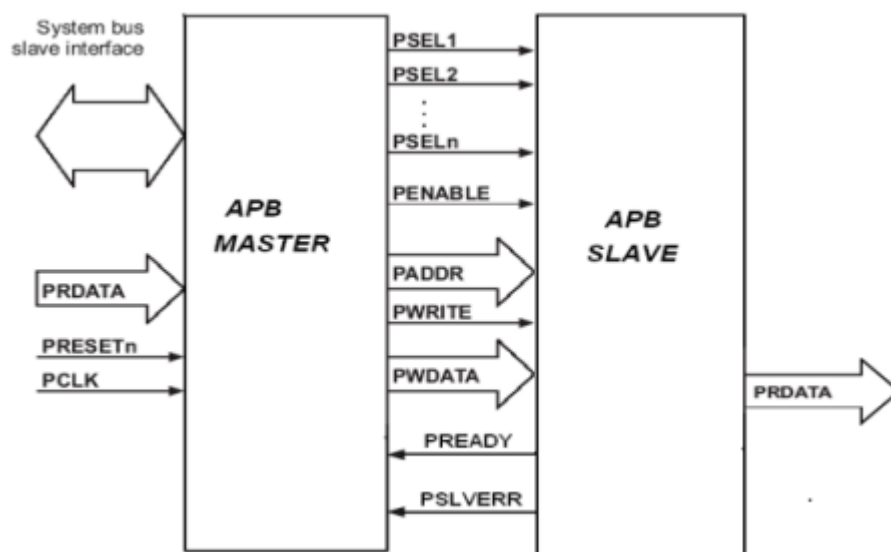


Figure 23 Interfacing between APB Master and Slave

2.2.1 Working of APB Protocol

a) Write Transfer – without wait states

At T1, a write transfer with address PADDR,PWDATA,PWRITE and PSEL starts. They will be registered at the next rising edge of PCLK, T2. This is Setup Phase of Transfer. After T2, PENABLE and PREADY are registered at the rising edge of PCLK. When asserted, PENABLE indicates starting of ACCESS Phase. When asserted, PREADY indicates that slave can complete the transfer at the next rising edge of PCLK. PADDR, PDATA and control signals all should remain valid till the transfer completes at T3. PENABLE signal will be de-asserted at the end of transfer. PSEL is also de-asserted, if next transfer is not to the same slave.

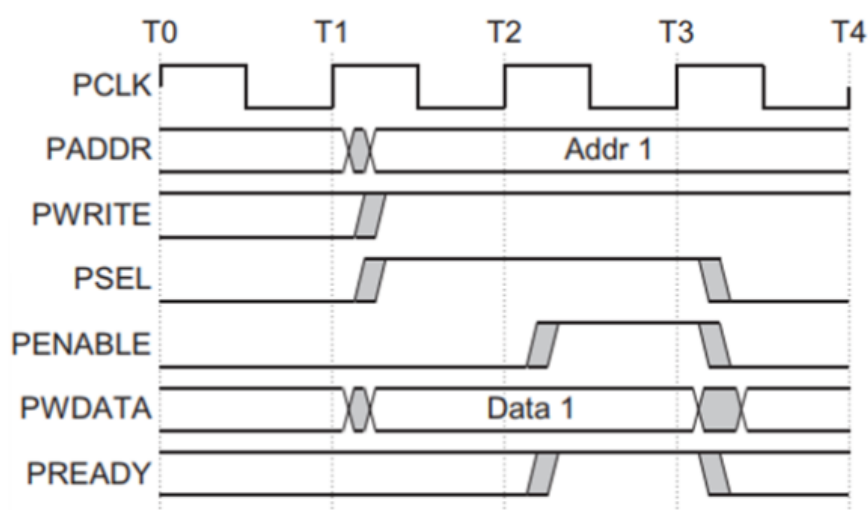


Figure 24 write transfer without wait states

b) Write Transfer – with Wait States

During the ACCESS Phase, when PENABLE is high, the slave extends the transfer by driving PREADY low. The PADDR, PWRITE, PSEL, PENABLE, PWDATA, PSTRB, PPROT signals should remain unchanged while PREADY is low. PREADY can take any value when PENABLE is low. It is recommended that the address and write signals are not changed immediately after a transfer, but remain stable until another access occurs.

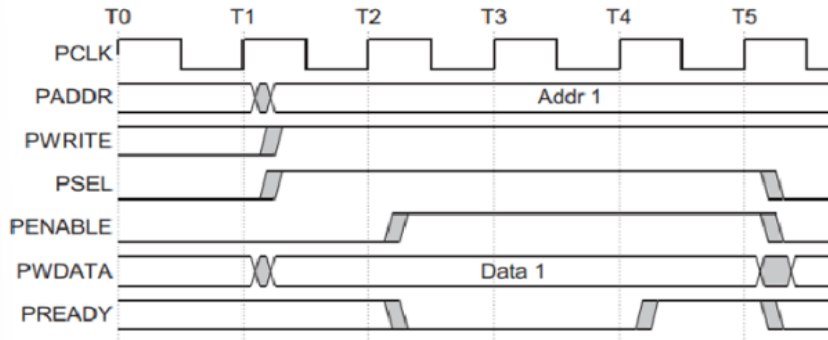


Figure 25 Write Transfer with wait states

c) Read Transfer- without wait states

At T1, a READ transfer with address PADDR, PWRITE and PSEL starts. They will be registered at rising edge of PCLK. This is SETUP Phase of the transfer. After T2, PENABLE and PREADY are registered at the rising edge of PCLK. When asserted, PENABLE indicates the starting of ACCESS phase. When asserted, PREADY indicates that slave can complete the transfer at next rising edge of PCLK by providing the data on PRDATA. Slave must provide the data before the end of read transfer. i.e. before T3.

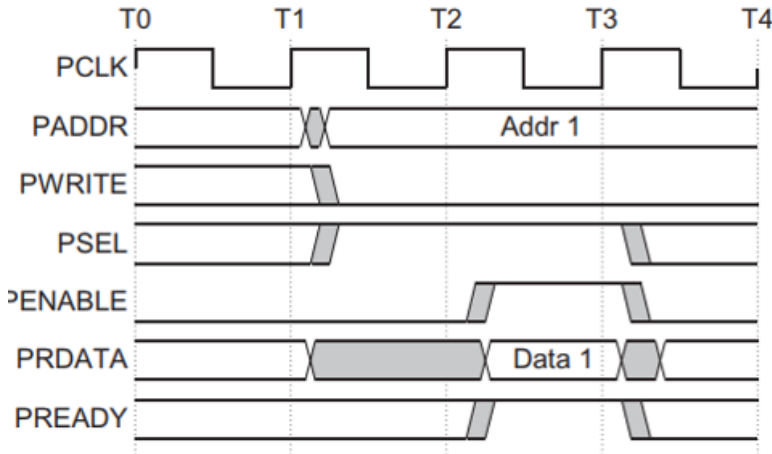


Figure 26 Read Transfer without Wait States

d) Read Transfer -with Wait States

During the ACCESS Phase, when PENABLE is high, the slave extends the transfer by driving PREADY low. The PADDR, PWRITE, PSEL, PENABLE, PPROT signals should remain unchanged while PREADY is low.

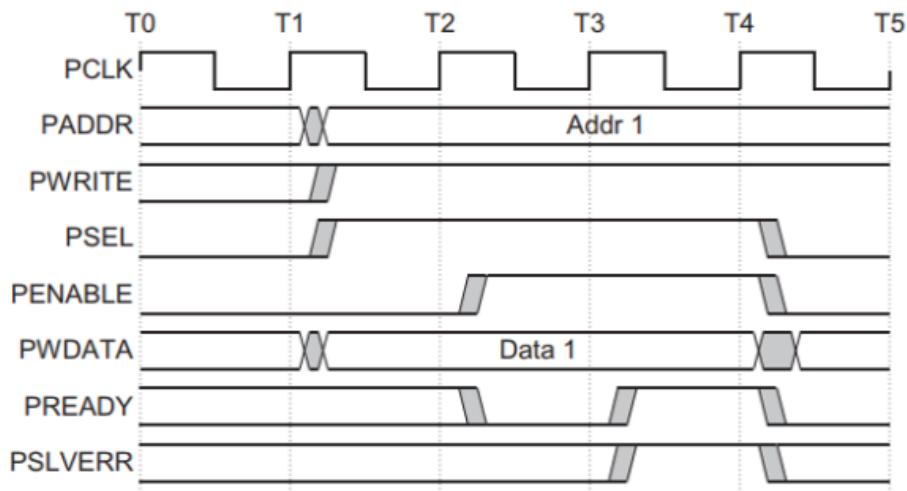


Figure 27 Read Transfer with wait States

e) Error Response

Whenever there is a problem in the transfer, Slave indicates the error response for the transfer by asserting the PSLVERR signal. PSLVERR is only considered valid during the last cycle of an APB transfer, when PSEL, PENABLE and PREADY are all HIGH. It is recommended, but not mandatory that you drive PSLVERR low when it is not being sampled.

Transactions that receive an error response, might or might not have changed the state of peripheral. For example, If APB master performs a write transaction to an APB slave and received an error response, it is not guaranteed that the data is not written on the slave peripheral.

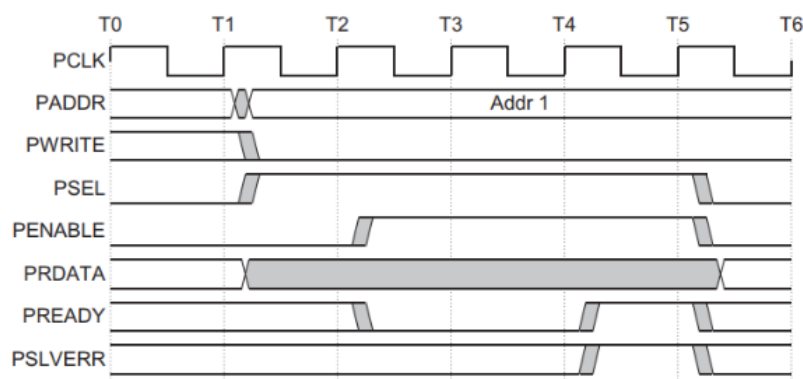


Figure 28 Error Response for Read Transfer

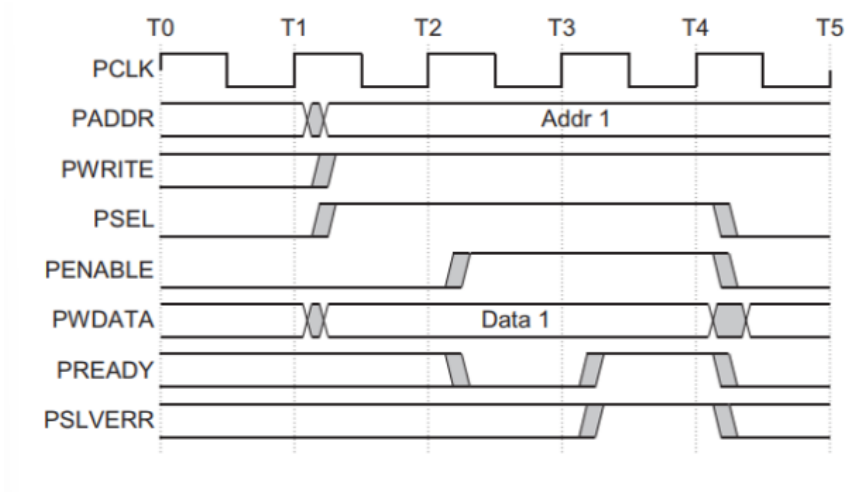


Figure 29 Error Response for Write Transfer

a) Protection Unit Support:

To support complex system designs, it is often necessary for both the interconnect and other devices in the system to provide protection against illegal transactions. It is provided by Protection Unit in APB Protocol. The signals indicating the protection unit are PPROT[2:0].

The three levels of access protection are

- **PPROT[0]:**
 - LOW indicates Normal Access
 - HIGH indicates Privileged Access
- **PPROT[1]:**
 - LOW indicates Secure Access
 - HIGH indicates Non-Secure Access
- **PPROT[2]:**
 - LOW indicates Data Access
 - HIGH indicates Instruction Access

2.3.10 Operating State Diagram

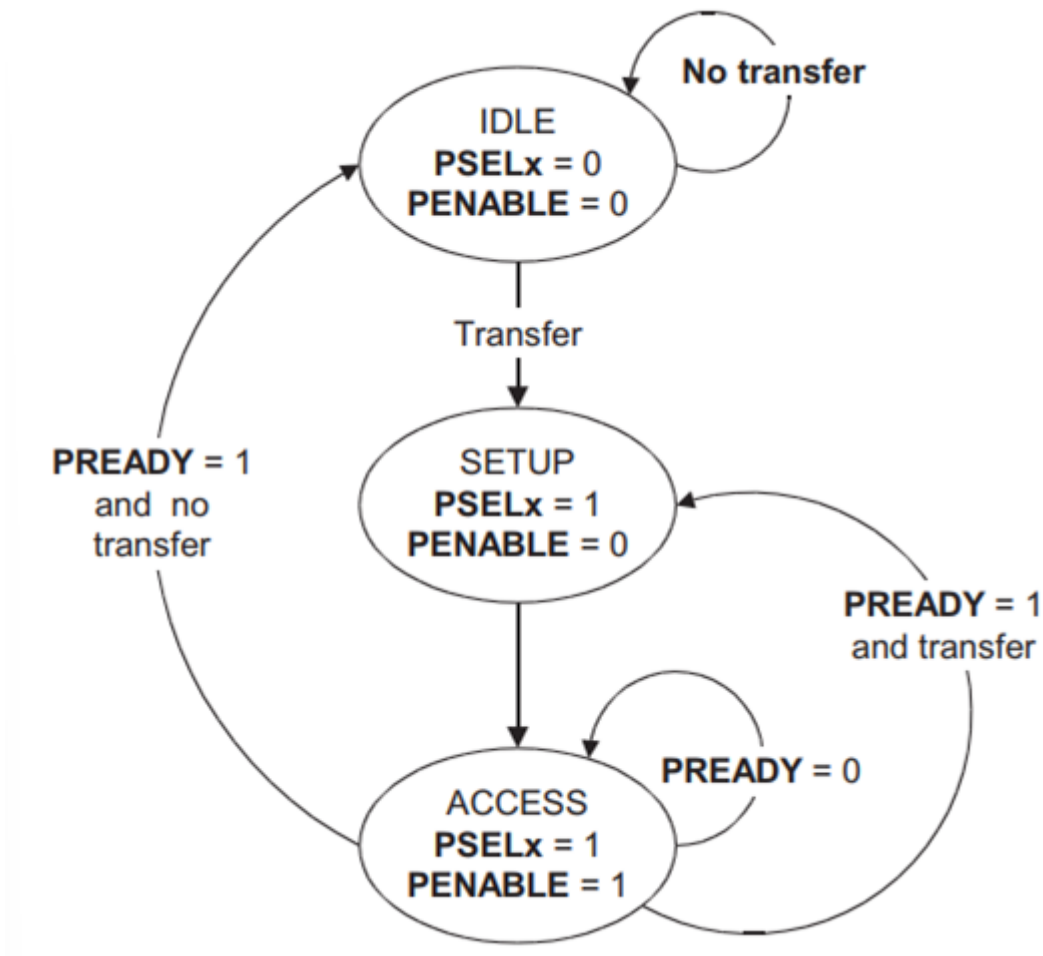


Figure 30 APB FSM

IDLE : This is the default state of APB.

SETUP : When transfer is required, PSELx is asserted then the bus moves in setup state. Bus only remains in SETUP for only one clock cycle and always moves to ACCESS state on next rising edge of clock. So, the slave must be able to sample the Address and control information in the SETUP cycle itself.

ACCESS : PENABLE is asserted to enter into the ACCESS state. The PADDR, PWRITE, PSELx and PWDATA signals must remain stable during ACCESS state.